A First Course in Abstract Algebra

Part 3: Homomorphisms and Factor Groups

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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13 Homomorphisms

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 13.1: Inclusion of the integers in the reals

Yes. Both groups use addition, so the required identity is $\phi(m+n) = \phi(m) + \phi(n)$. The left side is $m+n$ and the right side is also $m+n$. Thus the inclusion preserves the operation and is a homomorphism.

Exercise 13.2: The greatest-integer map

No. Choose $x = y = 3/5$. Then $\phi(x) = \phi(y) = 0$, but $\phi(x+y) = \lfloor 6/5 \rfloor = 1$. Hence $\phi(x+y) \neq \phi(x) + \phi(y)$, so the additive homomorphism law fails.

Exercise 13.3: Absolute value under multiplication

Yes. Absolute value sends nonzero reals to nonzero reals, so the map has the stated codomain. For every $x, y \in \mathbb{R}^*$,

$$\phi(xy) = |xy| = |x||y| = \phi(x)\phi(y).$$

The kernel is $\{1, -1\}$, showing that a homomorphism need not be injective.

Exercise 13.4: Reduction from $\mathbb{Z}_6$ to $\mathbb{Z}_2$

Yes. Reduction of an integer modulo 2 is unchanged if a multiple of 6 is added, because 6 is even. Thus reduction descends to $\mathbb{Z}_6$, and reducing the sum of two representatives modulo 2 gives the sum of their reductions. Formally $\phi([a+b]_6) = [a+b]_2 = [a]_2 + [b]_2$, proving the homomorphism law.

Exercise 13.5: Remainder parity on $\mathbb{Z}_9$

No. Using the canonical representatives, take 8 and 1. In $\mathbb{Z}_9$ their sum is 0, so $\phi(8+1) = \phi(0) = 0$. But $\phi(8) + \phi(1) = 0 + 1 = 1$ in $\mathbb{Z}_2$. Subtracting the modulus 9 changes parity, causing the homomorphism law to fail.

Exercise 13.6: Exponentiation from addition to multiplication

Yes. The values 2^x are positive and hence nonzero reals. The domain operation is addition and the target operation is multiplication, so the correct identity is

$$\phi(x+y) = 2^{x+y} = 2^x 2^y = \phi(x)\phi(y).$$

The change in operations is part of the homomorphism definition and must be kept explicit.

Exercise 13.7: A coordinate inclusion into a direct product

Yes. The map puts g_i in its own coordinate and identity elements in all others. Multiplying $\phi_i(a)$ and $\phi_i(b)$ multiplies only the ith coordinates nontrivially, giving ab there. Consequently $\phi_i(a)\phi_i(b) = \phi_i(ab)$. It is also injective because equality of the product vectors forces equality of their ith coordinates.

Exercise 13.8: The inverse map

For an arbitrary group, the answer depends on commutativity. One has

$$\phi(ab) = (ab)^{-1} = b^{-1}a^{-1}, \qquad \phi(a)\phi(b) = a^{-1}b^{-1}.$$

Thus it is a homomorphism if the group is abelian. Conversely, if it is a homomorphism, these expressions are equal for every a, b; taking inverses gives $ab = ba$, so the group must be abelian. For example it fails in S_3 using two noncommuting transpositions. The exact answer is: if and only if G is abelian.

Exercise 13.9: Taking a second derivative

Yes. The second derivative of a function having derivatives of all orders again has derivatives of all orders, so the map stays in F. Differentiation is additive, hence

$$\phi(f + g) = (f + g)'' = f'' + g'' = \phi(f) + \phi(g).$$

This is the homomorphism law for the additive function groups.

Exercise 13.10: Integration as an additive map

Yes. A continuous function has a finite Riemann integral on $[0, 4]$. Linearity of the integral gives

$$\phi(f + g) = \int_0^4 (f(x) + g(x))\,dx = \int_0^4 f(x)\,dx + \int_0^4 g(x)\,dx = \phi(f) + \phi(g).$$

Thus the map preserves the additive operations.

Exercise 13.11: Multiplying a function by three

Yes. For pointwise addition,

$$\phi(f + g) = 3(f + g) = 3f + 3g = \phi(f) + \phi(g).$$

The equality holds at every real input, so it is an equality of functions. There is no requirement here that the functions be continuous or differentiable.

Exercise 13.12: Determinant on an additive matrix group

For $n \geq 2$, no. The domain operation is matrix addition, so a homomorphism would require $\det(A + B) = \det A + \det B$. Taking $A = B = I_n$ gives $\det(2I_n) = 2^n$, while $\det I_n + \det I_n = 2$; these

are unequal for $n \geq 2$. The familiar multiplicative determinant identity concerns matrix multiplication and does not apply to this domain operation. For the exceptional case $n = 1$, the determinant is just the single entry, so the map is additive and is a homomorphism. Thus a blanket negative answer implicitly assumes $n > 1$.

Exercise 13.13: Trace on an additive matrix group

Yes. The diagonal entries of $A + B$ are $a_{ii} + b_{ii}$, so

$$\operatorname{tr}(A + B) = \sum_i (a_{ii} + b_{ii}) = \sum_i a_{ii} + \sum_i b_{ii} = \operatorname{tr}(A) + \operatorname{tr}(B).$$

This proves the additive homomorphism law for every positive matrix size.

Exercise 13.14: Trace on the multiplicative general linear group

No. A homomorphism from a multiplicative group to additive $\mathbb{R}$ must send the multiplicative identity to 0. But $\operatorname{tr}(I_n) = n \neq 0$ for $n \geq 1$. Equivalently, $I_n I_n = I_n$ would require $n = n + n$, which is false. The same formula that works on the additive matrix group fails for this different operation.

Exercise 13.15: Integration on a multiplicative function group

The map is well defined: a continuous nonvanishing function on $\mathbb{R}$ has constant sign, so its integral on $[0, 1]$ is nonzero. But it is not multiplicative. Take

$$f(x) = g(x) = 2 + \sin(2\pi x),$$

which is everywhere positive. Then $\phi(f) = \phi(g) = 2$, while

$$\phi(fg) = \int_0^1 \left(4 + 4\sin(2\pi x) + \sin^2(2\pi x)\right) dx = 4 + 0 + \tfrac{1}{2} = \tfrac{9}{2}.$$

This differs from $\phi(f)\phi(g) = 4$, so the homomorphism law fails.

Exercise 13.16: Kernel of the parity homomorphism

The map of Example 13.3 sends even permutations to $0 \in \mathbb{Z}_2$ and odd permutations to 1. The target identity is 0, so its inverse image is exactly the even permutations:

$$\operatorname{Ker} \phi = A_3 = \{e, (1, 2, 3), (1, 3, 2)\} = \{\rho_0, \rho_1, \rho_2\}.$$

Exercise 13.17: An integer homomorphism into $\mathbb{Z}_7$

Since 1 generates $\mathbb{Z}$ and $\phi(1) = 4$, additivity forces $\phi(n) = 4n \pmod 7$ for every integer n. Therefore $\phi(25) = 100 \equiv 2 \pmod 7$. The kernel consists of integers with $7 \mid 4n$. Since $\gcd(4, 7) = 1$, this is equivalent to $7 \mid n$, so $\operatorname{Ker} \phi = 7\mathbb{Z}$.

Exercise 13.18: An integer homomorphism into $\mathbb{Z}_{10}$

The image of n is $6n \pmod{10}$. Thus $\phi(18) = 108 \equiv 8 \pmod{10}$. Also $10 \mid 6n$ if and only if $5 \mid 3n$, which is equivalent to $5 \mid n$. Consequently $\operatorname{Ker}\phi = 5\mathbb{Z}$.

Exercise 13.19: An integer homomorphism into S_8

Let $g = (1,4,2,6)(2,5,7)$. Its factors overlap, so first compute their composition:

$$1 \mapsto 4 \mapsto 2 \mapsto 5 \mapsto 7 \mapsto 6 \mapsto 1, \qquad g = (1,4,2,5,7,6).$$

It has order 6, and $\phi(n) = g^n$, so $\operatorname{Ker}\phi = 6\mathbb{Z}$. Since $20 \equiv 2 \pmod{6}$,

$$\phi(20) = g^2 = (1,2,7)(4,5,6),$$

with the remaining symbols fixed.

Exercise 13.20: A homomorphism from $\mathbb{Z}_{10}$ to $\mathbb{Z}_{20}$

The formula is $\phi(n) = 8n \pmod{20}$, and it is consistent with the domain modulus because $8 \cdot 10 = 80 \equiv 0 \pmod{20}$. The kernel condition $20 \mid 8n$ reduces to $5 \mid n$, so in $\mathbb{Z}_{10}$ the kernel is $\{0,5\}$. Finally $\phi(3) = 24 \equiv 4 \pmod{20}$.

Exercise 13.21: A cyclic homomorphism into S_8

Put $g = (2,5)(1,4,6,7)$. The disjoint cycles have lengths $2, 4$, so g has order 4. Hence $\phi(n) = g^n$ is well defined on $\mathbb{Z}_{24}$, since $4 \mid 24$. Its kernel consists of multiples of 4:

$$\operatorname{Ker}\phi = \{0,4,8,12,16,20\}.$$

Because $14 \equiv 2 \pmod{4}$, squaring the cycles gives

$$\phi(14) = (1,6)(4,7).$$

Exercise 13.22: A map from two integer coordinates to one

By additivity,

$$\phi(m,n) = m\phi(1,0) + n\phi(0,1) = 3m - 5n.$$

Thus $\phi(-3,2) = -9 - 10 = -19$. For the kernel, solve $3m = 5n$. Coprimality forces $m = 5k$ and then $n = 3k$, with $k \in \mathbb{Z}$. Therefore

$$\operatorname{Ker}\phi = \{(5k,3k) : k \in \mathbb{Z}\} = \langle (5,3) \rangle.$$

Exercise 13.23: A map between integer lattices

The generator images give

$$\phi(m,n) = m(2,-3) + n(-1,5) = (2m - n, -3m + 5n).$$

Therefore $\phi(4,6) = (8-6, -12+30) = (2,18)$. If the image is $(0,0)$, the first coordinate gives $n = 2m$, and the second then gives $-3m + 10m = 7m = 0$. Hence $m = n = 0$, so the kernel is $\{(0,0)\}$.

Exercise 13.24: A two-generator homomorphism to S_{10}

Write $a = (3,5)(2,4)$ and $b = (1,7)(6,10,8,9)$. Their supports are disjoint, so they commute, making the formula $\phi(m,n) = a^m b^n$ consistent with the abelian domain. The orders are 2 and 4. If $a^m b^n = e$, restriction to the support of a gives $2 \mid m$, and restriction to that of b gives $4 \mid n$. Conversely those divisibilities clearly suffice. Thus

$$\operatorname{Ker}\phi = 2\mathbb{Z} \times 4\mathbb{Z}.$$

For the specified point, $a^3 = a$ and $b^{10} = b^2 = (6,8)(10,9)$, so

$$\phi(3,10) = (3,5)(2,4)(6,8)(9,10).$$

Exercise 13.25: Homomorphisms of $\mathbb{Z}$ onto itself

Every homomorphism has the form $\phi(n) = kn$, where $k = \phi(1) \in \mathbb{Z}$. Its image is $k\mathbb{Z}$. This is all of $\mathbb{Z}$ precisely when $k = 1$ or $k = -1$. Thus there are exactly two surjective homomorphisms, $n \mapsto n$ and $n \mapsto -n$.

Exercise 13.26: Homomorphisms of $\mathbb{Z}$ into itself

For each integer k, the formula $\phi_k(n) = kn$ is additive, so is a homomorphism. Conversely additivity and the generator 1 force this formula for every homomorphism, including negative inputs by $\phi(-n) = -\phi(n)$. Therefore the homomorphisms are parametrized by $k \in \mathbb{Z}$: countably infinitely many.

Exercise 13.27: Homomorphisms of $\mathbb{Z}$ into $\mathbb{Z}_2$

The image of 1 is either 0 or 1, and that choice determines the map on every integer. Both choices work: $n \mapsto 0$ and $n \mapsto n \bmod 2$ preserve addition. Thus there are exactly two homomorphisms.

Exercise 13.28: When is left multiplication a homomorphism?

A homomorphism must send identity to identity. Here $\phi_g(e) = ge = g$, so this requires $g = e$. Conversely, when $g = e$, ϕ_g is the identity map and is a homomorphism. Thus precisely the identity element qualifies. Being a permutation of G is much weaker than being a group homomorphism.

Exercise 13.29: When is conjugation a homomorphism?

For every $g \in G$. Indeed

$$\phi_g(xy) = gxyg^{-1} = (gxg^{-1})(gyg^{-1}) = \phi_g(x)\phi_g(y),$$

where the middle $g^{-1}g$ cancels. Its inverse map is conjugation by g^{-1}, so it is even an automorphism for every g.

Exercise 13.30: Definition of a group homomorphism

A complete definition specifies the algebraic structures and the quantifier: a homomorphism from a group G to a group G' is a function $\phi : G \to G'$ satisfying $\phi(xy) = \phi(x)\phi(y)$ for all $x, y \in G$. The first product is in G and the second is in G'. Merely saying "a map" without domains, operations, and a universal condition leaves the original statement incomplete.

Exercise 13.31: Definition of the kernel

The definition is correct. For a group homomorphism $\phi : G \to G'$ with target identity e', the kernel is the inverse image of that identity:

$$\operatorname{Ker} \phi = \{x \in G : \phi(x) = e'\}.$$

It always contains the identity of G and is a normal subgroup; the identity used in the test is the target identity, which need not have the same notation or value as the domain identity.

Exercise 13.32: True or false

(a) True: A_n is the kernel of the sign homomorphism (with the trivial small cases included).

(b) True: the constant map $g \mapsto e'$ is always a homomorphism.

(c) False: a nontrivial group mapping constantly to the target identity has many points in its kernel.

(d) True: $\phi(x) = \phi(y)$ is equivalent to $xy^{-1} \in \operatorname{Ker} \phi$, giving injectivity exactly for a trivial kernel.

(e) False: an image of a six-element group has order dividing 6, so cannot have order 4.

(f) False: a function on six points has at most six image points, so cannot have an image of size 12.

(g) True: $\mathbb{Z}_6 \to \mathbb{Z}_{12}$, $[k]_6 \mapsto [2k]_{12}$, is a homomorphism, even injective.

(h) True: at least the trivial homomorphism exists; a nontrivial example is $\mathbb{Z}_6 \to \mathbb{Z}_{10}$, $[k] \mapsto [5k]$.

(i) False: the domain identity always maps to the target identity, so the kernel is never empty.

(j) False: $\mathbb{Z}_2 \to \mathbb{R}^*$, $[k] \mapsto (-1)^k$, is a nontrivial homomorphism into an infinite group.

Exercise 13.33: A nontrivial map from $\mathbb{Z}_{12}$ to $\mathbb{Z}_5$?

No such map exists. Its image order must divide both the domain order 12 and the target order 5. Their gcd is 1, so the image has order 1 and the map is trivial. Equivalently, if $a = \phi(1)$, then $12a = 0$ in $\mathbb{Z}_5$ forces $a = 0$.

Exercise 13.34: A nontrivial map from $\mathbb{Z}_{12}$ to $\mathbb{Z}_4$

Use reduction modulo 4: $\phi([k]_{12}) = [k]_4$. It is well defined because changing k by 12 does not change its residue modulo 4, and it preserves addition. It is nontrivial because $\phi(1) = 1 \neq 0$, and is in fact surjective.

Exercise 13.35: A nontrivial map between two direct products

For the source $\mathbb{Z}_2 \times \mathbb{Z}_4$ and target $\mathbb{Z}_2 \times \mathbb{Z}_5$, define $\phi(a, b) = (a, 0)$. Addition is coordinatewise, so

$$\phi((a, b) + (c, d)) = (a + c, 0) = (a, 0) + (c, 0).$$

Thus it is a homomorphism. The value $\phi(1, 0) = (1, 0)$ is nonidentity, so it is nontrivial.

Exercise 13.36: A nontrivial map from $\mathbb{Z}_3$ to $\mathbb{Z}$?

Impossible. If $k = \phi(1)$, then $3k = \phi(3 \cdot 1) = \phi(0) = 0$ in the integers. This forces $k = 0$, so every element maps to zero. The obstruction is that a finite-order element cannot have a nonzero image in the torsion-free group $\mathbb{Z}$.

Exercise 13.37: A nontrivial map from $\mathbb{Z}_3$ to S_3

Let $r = (1, 2, 3)$ and define $\phi([k]_3) = r^k$. Since $r^3 = e$, this is independent of the representative. The exponent law gives $\phi(a + b) = r^{a+b} = r^a r^b = \phi(a)\phi(b)$. Its image is A_3, and $\phi(1) = r \neq e$, so it is nontrivial.

Exercise 13.38: A nontrivial map from $\mathbb{Z}$ to S_3

Take $\phi(k) = (1, 2)^k$. Integer powers satisfy $\phi(k + \ell) = \phi(k)\phi(\ell)$, including negative exponents. The map is nontrivial because $\phi(1) = (1, 2) \neq e$. Its image is the order-2 subgroup $\{e, (1, 2)\}$.

Exercise 13.39: A nontrivial map from $\mathbb{Z}^2$ to $2\mathbb{Z}$

Define $\phi(m, n) = 2m$. Its values lie in the even integers, and

$$\phi((m, n) + (r, s)) = 2(m + r) = 2m + 2r = \phi(m, n) + \phi(r, s).$$

It is nontrivial since $\phi(1, 0) = 2$, and is even onto $2\mathbb{Z}$.

Exercise 13.40: A nontrivial map from $2\mathbb{Z}$ to $\mathbb{Z}^2$

For an even integer $2k$, define $\phi(2k) = (k, 0)$. The integer k is unique, so the map is well defined. Then $\phi(2k + 2\ell) = (k + \ell, 0) = (k, 0) + (\ell, 0)$. It is nontrivial because $\phi(2) = (1, 0)$.

Exercise 13.41: A nontrivial map from D_4 to S_3

View D_4 as permutations of its four vertices. Define $\phi(g) = e$ when g is even and $\phi(g) = (1, 2)$ when g is odd. Parity adds modulo 2, and $(1, 2)^2 = e$, so this is a homomorphism into S_3. A quarter-turn is a 4-cycle and is odd, so it has nonidentity image, proving the map is nontrivial.

Exercise 13.42: A nontrivial map from S_3 to S_4

Extend a permutation of $\{1, 2, 3\}$ by fixing 4. The extension of a composite equals the composite of the extensions, because this is true on $1, 2, 3$ and all maps fix 4. Thus the extension is an injective homomorphism and is nontrivial.

Exercise 13.43: A nontrivial map from S_4 to S_3

Send each even permutation to $e \in S_3$ and each odd permutation to $(1, 2)$. This is the sign homomorphism with its two values relabeled by the order-2 subgroup $\{e, (1, 2)\}$ of S_3. Products preserve parity, so it preserves the group operation. It is nontrivial because a transposition has image $(1, 2)$. Surjectivity onto S_3 is not required by the exercise.

Exercise 13.44: Image order divides a finite domain order

Let $K = \operatorname{Ker}\phi$. For an image element $\phi(a)$, its full fiber is aK: indeed $\phi(x) = \phi(a)$ if and only if $\phi(a^{-1}x) = e'$, or $a^{-1}x \in K$. Each fiber therefore has $|K|$ elements by $k \mapsto ak$, and the fibers partition finite G. Thus

$$|G| = |\phi[G]|\,|K|.$$

The image is finite and its order divides $|G|$. This proof obtains the count directly without presupposing the isomorphism theorem of the next section.

Exercise 13.45: Image order divides a finite target order

The image $\phi[G]$ is a subgroup of G': it contains $e' = \phi(e)$, products satisfy $\phi(a)\phi(b) = \phi(ab)$, and inverses satisfy $\phi(a)^{-1} = \phi(a^{-1})$. Since G' is finite, its subgroup is finite and Lagrange gives $|\phi[G]| \mid |G'|$. The domain need not be finite for this argument.

Exercise 13.46: Agreement on generators determines a homomorphism

Every element of the group generated by the a_i is a finite word $a_{i_1}^{\varepsilon_1} \cdots a_{i_k}^{\varepsilon_k}$ with $\varepsilon_j = \pm 1$. A homomorphism carries this word to the product of the corresponding image powers, because it preserves products and inverses. Since $\phi(a_i) = \mu(a_i)$ for every generator, the two maps give equal products on every such word. They also agree at the identity. Thus they agree on every element of G, meaning $\phi = \mu$. No assumption that the generators are independent is needed.

Exercise 13.47: A homomorphism with prime-order domain

If $|G| = p$ is prime, its kernel is a subgroup and must have order 1 or p by Lagrange. In the first case the kernel is trivial, so the homomorphism is injective: equal images imply the quotient of the inputs lies in that kernel. In the second case the kernel is all of G, so every input maps to the target identity and the homomorphism is trivial. These are the only possibilities.

Exercise 13.48: Kernel of the sign map

By definition $\operatorname{sgn}(\sigma) = +1$ exactly for even permutations. The identity of the multiplicative target $\{1, -1\}$ is $+1$, so

$$\operatorname{Ker}(\operatorname{sgn}_n) = A_n.$$

For $n \geq 2$, a transposition has sign -1, so the map is onto both target values. For $n = 1$ it is still a homomorphism with kernel $S_1 = A_1$, but it is not onto the two-element target. This map is the general version of the parity map $S_3 \to \mathbb{Z}_2$.

Exercise 13.49: Composition of homomorphisms

For $a, b \in G$, apply first the homomorphism law for ϕ, then the one for ψ:

$$(\psi\phi)(ab) = \psi(\phi(ab)) = \psi(\phi(a)\phi(b)) = \psi(\phi(a))\psi(\phi(b)).$$

The result is $(\psi\phi)(a)(\psi\phi)(b)$, which is the required law for the composite $G \to G''$.

Exercise 13.50: An abelian image and commutators

For any $x, y \in G$,

$$\phi(xyx^{-1}y^{-1}) = \phi(x)\phi(y)\phi(x)^{-1}\phi(y)^{-1}.$$

If the image is abelian, the right side cancels to the identity, so every displayed commutator lies in the kernel. Conversely, if it is always in the kernel, the right side equals the identity. Multiplying that equality on the right by $\phi(y)\phi(x)$ gives $\phi(x)\phi(y) = \phi(y)\phi(x)$. Every image element is some $\phi(x)$, so the image is abelian.

Exercise 13.51: The universal power map from the integers

For all integers m, n, $a^{m+n} = a^m a^n$, so $\phi(n) = a^n$ is a homomorphism. Its image is exactly $\langle a \rangle$. If a has infinite order, only the exponent 0 gives e, so the kernel is $\{0\}$. If a has finite order r, then $a^n = e$ exactly when $r \mid n$, so the kernel is $r\mathbb{Z}$. This includes $r = 1$, when $a = e$ and the kernel is all of $\mathbb{Z}$.

Exercise 13.52: A fiber is a kernel coset

Let $H = \operatorname{Ker}\phi$. For $x \in G$,

$$\phi(x) = \phi(a) \iff \phi(xa^{-1}) = e' \iff xa^{-1} \in H \iff x \in Ha.$$

The first equivalence uses preservation of inverses and cancellation in the target; the last is the definition of a right coset. Thus the two described sets are equal. Since a kernel is normal, this fiber is also aH.

Exercise 13.53: A two-generator power formula

The necessary and sufficient condition is $hk = kh$. If $\phi(m, n) = h^m k^n$ is a homomorphism, the commuting domain elements $(1, 0)$ and $(0, 1)$ must have commuting images. Their images are h, k,

proving necessity. Conversely suppose h, k commute. Their integer powers then commute, including negative powers, and for all integers m, n, r, s,

$$\phi(m,n)\phi(r,s) = h^m k^n h^r k^s = h^{m+r} k^{n+s} = \phi(m+r, n+s).$$

This is the additive-domain homomorphism law, proving sufficiency.

Exercise 13.54: When all two-generator power formulas work

The condition is that G be abelian. If every pair h, k gives a homomorphism in Exercise 53, its necessary condition says every pair commutes, so G is abelian. Conversely, if G is abelian, every pair commutes and the sufficient calculation in that exercise proves every such formula is a homomorphism.

Exercise 13.55: Power maps from a finite cyclic group

The exact condition is

$$\boxed{h^n = e,}$$

equivalently, h has finite order dividing n. Necessity follows because $n[1] = [0]$ in $\mathbb{Z}_n$, so a homomorphism must satisfy $h^n = \phi(n[1]) = \phi([0]) = e$. For sufficiency, if integers i, j differ by a multiple of n, then $h^i = h^j$ because $h^n = e$. Thus $\phi([i]) = h^i$ is well defined on residue classes, and

$$\phi([i] + [j]) = h^{i+j} = h^i h^j = \phi([i])\phi([j]).$$

The source writes $0 \leq i \leq n$, duplicating $[0] = [n]$; the same condition resolves that endpoint consistency. It requires $h^n = e$, not $h = e$: for $n = 2$ and $G = \mathbb{Z}_2$, sending 1 to the nonzero element gives the identity homomorphism and has $h \neq e$.

14 Factor Groups

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 14.1: Order of $\mathbb{Z}_6/\langle 3 \rangle$

The subgroup is $\langle 3 \rangle = \{0, 3\}$, of order 2. The quotient order is the number of cosets, $6/2 = 3$. Explicitly the cosets are $\{0, 3\}$, $\{1, 4\}$, $\{2, 5\}$.

Exercise 14.2: Quotient by a product subgroup

In $\mathbb{Z}_4$, $\langle 2 \rangle$ has order 2; in $\mathbb{Z}_{12}$, $\langle 2 \rangle$ has order 6. Their product therefore has order $2 \cdot 6 = 12$. The ambient product has order $4 \cdot 12 = 48$, so the quotient order is $48/12 = 4$. All groups here are abelian, so the subgroup is normal and the quotient is defined.

Exercise 14.3: Quotient by $\langle (2, 1) \rangle$

In $\mathbb{Z}_4 \times \mathbb{Z}_2$, the coordinates 2 and 1 each have order 2, so $(2, 1)$ has order 2 and generates $\{(0, 0), (2, 1)\}$. The ambient order is $4 \cdot 2 = 8$. Thus the quotient has order $8/2 = 4$.

Exercise 14.4: Quotient by the second coordinate factor

The subgroup $\{0\} \times \mathbb{Z}_5$ has five elements in the fifteen-element group $\mathbb{Z}_3 \times \mathbb{Z}_5$, so the quotient has order 3. Equivalently the projection $(a, b) \mapsto a$ has this subgroup as kernel and image $\mathbb{Z}_3$.

Exercise 14.5: Quotient of $\mathbb{Z}_2 times \mathbb{Z}_4$ by a diagonal cycle

The element $(1, 1)$ has order $\operatorname{lcm}(2, 4) = 4$, so its generated subgroup has four elements. The product has 8 elements, giving a quotient of order $8/4 = 2$.

Exercise 14.6: Quotient of $\mathbb{Z}_{12} times \mathbb{Z}_{18}$

The order of $4 \in \mathbb{Z}_{12}$ is $12/\gcd(12, 4) = 3$, and that of $3 \in \mathbb{Z}_{18}$ is $18/\gcd(18, 3) = 6$. Thus $(4, 3)$ has order $\operatorname{lcm}(3, 6) = 6$. Dividing the ambient order $12 \cdot 18 = 216$ by this subgroup order gives quotient order $216/6 = 36$.

Exercise 14.7: A quotient of $\mathbb{Z}_2 times S_3$

The element $(1, \rho_1)$ has order $\operatorname{lcm}(2, 3) = 6$. Its subgroup equals $\mathbb{Z}_2 \times A_3$: both have six elements, and all its powers lie there. This is normal because $A_3 \triangleleft S_3$ and the first coordinate is unrestricted. The ambient order is $2 \cdot 6 = 12$, so the quotient has order $12/6 = 2$. Normality matters here because the ambient group is not abelian.

Exercise 14.8: A diagonal subgroup of a coprime product

The element $(1,1)$ in $\mathbb{Z}_{11} \times \mathbb{Z}_{15}$ has order $\operatorname{lcm}(11,15) = 165$, equal to the ambient order. It therefore generates the whole group, so the quotient has just one element. Its order is 1.

Exercise 14.9: Order of $5 + \langle 4 \rangle$ in a cyclic quotient

The subgroup of $\mathbb{Z}_{12}$ is $H = \{0, 4, 8\}$. The order of $5 + H$ is the least positive k with $5k \in H$. Testing multiples modulo 12 gives $5, 10, 3, 8$ at $k = 1, 2, 3, 4$, respectively; the first three are outside H and 8 is inside. Thus the order is 4. Equivalently reduce to $5 \equiv 1$ modulo 4.

Exercise 14.10: Order of $26 + \langle 12 \rangle$

The quotient $\mathbb{Z}_{60}/\langle 12 \rangle$ is identified with $\mathbb{Z}_{12}$ by reduction modulo 12, because the kernel consists exactly of the multiples of 12 modulo 60. The coset of 26 corresponds to $2 \in \mathbb{Z}_{12}$, whose order is $12/\gcd(12,2) = 6$. Thus the requested order is 6.

Exercise 14.11: Order of a coset in $\mathbb{Z}_3 times \mathbb{Z}_6$

Let $H = \langle (1,1) \rangle$. Its elements are $(t \bmod 3, t \bmod 6)$. A multiple $k(2,1)$ lies in H if and only if an integer t satisfies

$$t \equiv 2k \pmod 3, \qquad t \equiv k \pmod 6.$$

The second congruence implies $t \equiv k \pmod 3$, so consistency is exactly $3 \mid k$. This condition is also sufficient by taking $t = k$. Therefore the least positive such k is 3, giving order 3.

Exercise 14.12: Order of a coset in $\mathbb{Z}_4^2$

The homomorphism $(x, y) \mapsto x - y \pmod 4$ has kernel $\langle (1,1) \rangle$, so the coset of $(3,1)$ corresponds to $3 - 1 = 2 \in \mathbb{Z}_4$. This element has order 2. Directly, $(3,1)$ is not on the diagonal, but $2(3,1) = (2,2)$ is, giving the same least positive return.

Exercise 14.13: Order modulo $\langle (0,2) \rangle$

In $\mathbb{Z}_4 \times \mathbb{Z}_8$, the subgroup consists of $(0,0), (0,2), (0,4), (0,6)$. For $k(3,1)$ to lie in it, the first coordinate must vanish, requiring $4 \mid 3k$ or $4 \mid k$, and the second coordinate must be even, requiring $2 \mid k$. The combined least positive solution is $k = 4$. Hence the coset has order 4.

Exercise 14.14: Order modulo $\langle (1,2) \rangle$

In $\mathbb{Z}_4 \times \mathbb{Z}_8$, use $f(x,y) = y - 2x \pmod 8$. This is well defined: changing x by 4 changes $-2x$ by -8. Its kernel is exactly $\{(t, 2t) : t \in \mathbb{Z}_4\} = \langle (1,2) \rangle$. Thus the coset of $(3,3)$ has the same order as $f(3,3) = 3 - 6 = -3 \equiv 5 \pmod 8$. Since $\gcd(5,8) = 1$, its order is 8. This also verifies that no smaller positive multiple enters the subgroup.

Exercise 14.15: A coset already equal to the identity

In $\mathbb{Z}_6 \times \mathbb{Z}_8$ one has $2(4,4) = (8,8) = (2,0)$. Hence $(2,0)$ already belongs to $H = \langle (4,4) \rangle$. The coset $(2,0) + H$ is therefore H, the identity of the quotient, and its order is 1.

Exercise 14.16: Conjugating a reflection subgroup of S_3

The source convention is $i_g(x) = gxg^{-1}$. Here $\rho_1 = (1,2,3)$ and $\mu_1 = (2,3)$. Conjugation relabels a transposition by applying the conjugating permutation to its entries:

$$\rho_1(2,3)\rho_1^{-1} = (\rho_1(2), \rho_1(3)) = (3,1) = \mu_2.$$

The identity remains the identity, so $i_{\rho_1}[H] = \{\rho_0, \mu_2\}$.

Exercise 14.17: Correcting the first normal-subgroup definition

The definition must read: a subgroup $H \leq G$ is normal if $gH = Hg$ for every $g \in G$. The original condition $hG = Gh$ for $h \in H$ says nothing about normality, because for any $h \in G$ both hG and Gh are just the whole group G. The subgroup being translated must be H, and the quantifier must range over all of G.

Exercise 14.18: The conjugation definition of normality

The stated definition is correct, provided H is a subgroup: $g^{-1}hg \in H$ for every $h \in H$ and $g \in G$ characterizes normality. Although this initially states an inclusion $g^{-1}Hg \subseteq H$, replacing g by g^{-1} gives the reverse inclusion after conjugation, so equality follows. Thus it is equivalent to invariance under every inner automorphism.

Exercise 14.19: Definition of an automorphism

An automorphism of G is an *isomorphism* from G onto itself, meaning a bijective homomorphism $G \to G$. A homomorphism merely mapping G into itself is an endomorphism and need not be invertible. For example, the constant map to e is an endomorphism of every group but is not an automorphism unless the group is trivial.

Exercise 14.20: Why normal subgroups matter

Normality is exactly the condition making the rule $(aH)(bH) = (ab)H$ independent of the chosen representatives, so that the cosets form the factor group G/H. Every normal subgroup is the kernel of the canonical homomorphism $g \mapsto gH$, and conversely every kernel is normal. Thus normal subgroups allow us to collapse specified elements to the identity while retaining a valid group operation, and describe the information lost by homomorphisms.

Exercise 14.21: Writing a proof about elements of a quotient

(a) The instructor suspects that the writer is confusing elements of G with elements of G/H, which are cosets. Naming quotient elements a, b is formally permissible, but then one cannot invoke commutativity of G on those objects without choosing representatives. The difficulty is not the letter names themselves; it is failure to track which group they belong to. **(b)** A clear beginning is: "Let $aH, bH \in G/H$, where $a, b \in G$. We must show $(aH)(bH) = (bH)(aH)$." **(c)** Since G is abelian, $ab = ba$. Therefore

$$(aH)(bH) = (ab)H = (ba)H = (bH)(aH).$$

The quotient multiplication is valid because H is normal. This proves G/H is abelian.

Exercise 14.22: A quotient of a torsion group

(a) As in Exercise 21, the likely error is treating a coset as though it were an element of the original group. The sentence "let $x \in G/H$" is not itself false, but the torsion hypothesis is about representatives in G, so the proof must display one. **(b)** Write: "Let $aH \in G/H$ with $a \in G$. Since G is torsion, a has finite order r." **(c)** Then $(aH)^r = a^r H = eH = H$, the quotient identity. Thus aH has finite order dividing r. Since the coset was arbitrary, the quotient is torsion. Its order need not equal the representative's order; it can decrease.

Exercise 14.23: True or false

(a) True: the standard coset product defines a group exactly when $N \triangleleft G$.

(b) True: $ghg^{-1} = h$ in an abelian group.

(c) True: conjugating x by g gives $gxg^{-1} = x$ when the group is abelian.

(d) True: its order is $|G|/|H|$.

(e) True: if $a^r = e$, then $(aH)^r = H$.

(f) False: $\mathbb{Z}$ is torsion free but $\mathbb{Z}/2\mathbb{Z}$ has a nonidentity element of order 2.

(g) True: $(aH)(bH) = abH = baH = (bH)(aH)$.

(h) False: $S_3/A_3 \cong \mathbb{Z}_2$ is abelian, although S_3 is not.

(i) True for positive integer n: the class of 1 generates all n residue classes.

(j) False in general: for nonzero integer n, $n\mathbb{R} = \mathbb{R}$, since $x = n(x/n)$, so the quotient is trivial, not of order n unless $n = 1$.

Exercise 14.24: The alternating subgroup and its quotient

For $n \geq 2$, the sign map $\operatorname{sgn} : S_n \to \{1, -1\}$ is a surjective homomorphism with kernel A_n. A kernel is normal, so $A_n \triangleleft S_n$, and the homomorphism theorem gives

$$S_n/A_n \cong \{1, -1\} \cong \mathbb{Z}_2.$$

Equivalently there are two cosets, the even and the odd permutations; odd times odd is even. For $n = 1$, $S_1 = A_1$ and the quotient is trivial, which is the small-case exception to the order-2 conclusion.

Exercise 14.25: Well-defined coset multiplication forces an inclusion

Let $h \in H$. Since $hH = H = eH$, well-definedness of multiplication gives

$$(ha)H = (hH)(aH) = (eH)(aH) = aH.$$

The element ha lies in its own left coset $(ha)H$, so $ha \in aH$. Since this holds for every $h \in H$, it proves $Ha \subseteq aH$. This is the desired missing direction and uses equality of cosets rather than an unproved normality assumption.

Exercise 14.26: Quotienting out all torsion

The torsion elements form a subgroup T by the earlier proof for abelian groups, and every subgroup of an abelian group is normal. Suppose gT has finite order in G/T. Then $(gT)^n = T$ for some positive n, so $g^n \in T$. Because that element is torsion, $(g^n)^m = e$ for some positive m. Thus $g^{nm} = e$, so $g \in T$ and $gT = T$. The quotient has no nonidentity torsion elements, which means it is torsion free.

Exercise 14.27: Conjugacy is an equivalence relation on subgroups

Reflexivity follows from $H = eHe^{-1}$. For symmetry, if $K = gHg^{-1}$ then $H = g^{-1}Kg$. For transitivity, if $K = gHg^{-1}$ and $L = hKh^{-1}$, then

$$L = hgHg^{-1}h^{-1} = (hg)H(hg)^{-1}.$$

Thus conjugacy of subgroups is reflexive, symmetric, and transitive. Each conjugation maps subgroups to subgroups, since it is an automorphism.

Exercise 14.28: Normal subgroups as conjugacy classes

A subgroup H is normal exactly when its conjugacy class among subgroups is the singleton $\{H\}$. Indeed, normality means $gHg^{-1} = H$ for every g, so there are no other conjugate subgroups. Conversely a singleton class means every conjugate equals H, which is the conjugation criterion for normality. This is a class of subgroups, not a conjugacy class of individual group elements.

Exercise 14.29: Conjugate reflection subgroups in S_3

The subgroup $\{\rho_0, \mu_2\}$ is generated by the transposition $(1, 3)$. Conjugating a transposition gives $(g(1), g(3))$, and as g ranges over S_3 this realizes every transposition. Hence the conjugate subgroups are exactly

$$\{\rho_0, \mu_1\}, \quad \{\rho_0, \mu_2\}, \quad \{\rho_0, \mu_3\}.$$

There are no others, since any conjugate subgroup has two elements and its nonidentity element is a transposition.

Exercise 14.30: Powers lying in a normal subgroup of finite index

Interpret $m = [G : H]$ as finite, as required for the integer exponent in the statement. The factor group has order m, so Lagrange gives $(aH)^m = H$ for each coset. Since $(aH)^m = a^m H$, this is $a^m H = H$, equivalent to $a^m \in H$. The ambient group itself need not be finite; finite index is enough. If the index is infinite, the displayed power a^m is not an ordinary group power and the statement needs the finite-index qualification.

Exercise 14.31: Intersections of normal subgroups

Let $N = \bigcap_{i \in I} N_i$ with each $N_i \triangleleft G$. It is a subgroup: all identities belong, and if $a, b \in N$, then ab^{-1} belongs to every N_i, hence to N. For $g \in G$ and $x \in N$, normality of every N_i gives $gxg^{-1} \in N_i$ for all i, so $gxg^{-1} \in N$. Thus N is normal. For an empty family the intersection is understood to be G, which is also normal.

Exercise 14.32: The normal closure of a subset

Let $\mathcal{N}$ be the collection of normal subgroups containing S. It is nonempty because G itself belongs to it. Set

$$N_S = \bigcap_{N \in \mathcal{N}} N.$$

By Exercise 31 this is normal. It contains S, since every intersected subgroup does. If M is any normal subgroup containing S, then $M \in \mathcal{N}$ and $N_S \subseteq M$ by the definition of intersection. Thus it is the unique smallest normal subgroup containing S, called the normal closure.

Exercise 14.33: The commutator quotient is abelian

By definition every commutator $aba^{-1}b^{-1}$ lies in the normal subgroup C. Therefore in G/C,

$$(aC)(bC)(aC)^{-1}(bC)^{-1} = (aba^{-1}b^{-1})C = C.$$

Multiplying by $(bC)(aC)$ on the right gives $(aC)(bC) = (bC)(aC)$. All quotient elements are such cosets, so the quotient is abelian. This proof uses only the inclusion of all commutators, not that each element of C must itself be a single commutator.

Exercise 14.34: A unique subgroup of a given order is normal

For any $g \in G$, conjugation is an automorphism, so gHg^{-1} is a subgroup of the same order as H. If H is the only subgroup of that order, $gHg^{-1} = H$ for every g. This is exactly normality. Finiteness ensures that "order" in the premise is the usual integer cardinality, although the uniqueness argument itself works for arbitrary cardinality.

Exercise 14.35: Intersecting with a normal subgroup

For $h \in H$ and $x \in H \cap N$, the conjugate hxh^{-1} lies in H because H is a subgroup, and in N because $N \triangleleft G$. Thus it lies in $H \cap N$, proving $H \cap N \triangleleft H$. It need not be normal in G: take $G = N = S_3$ and

$H = \{e, (1,2)\}$. Then N is normal in G and $H \cap N = H$. But conjugating $(1,2)$ by $(1,2,3)$ gives $(2,3) \notin H$, so this intersection is not normal in G.

Exercise 14.36: Intersection of all subgroups of a fixed finite order

Let K be the intersection of all subgroups of order s; the family is nonempty by hypothesis. Conjugation by a fixed $g \in G$ permutes that family, since $H \mapsto gHg^{-1}$ preserves subgroup order and has inverse conjugation by g^{-1}. Thus

$$gKg^{-1} = g\left(\bigcap_{|H|=s} H\right)g^{-1} = \bigcap_{|H|=s} gHg^{-1} = \bigcap_{|H|=s} H = K.$$

Conjugation commutes with intersection because it is a bijection. Since an intersection of subgroups is a subgroup, the equality proves $K \triangleleft G$ even though the individual subgroups need not be normal.

Exercise 14.37: Automorphisms and inner automorphisms

(a) The identity is an automorphism. A composite of two bijective homomorphisms is again a bijective homomorphism. The inverse of a bijective homomorphism is a homomorphism: writing $x = \phi(a)$, $y = \phi(b)$ gives $\phi^{-1}(xy) = \phi^{-1}(\phi(ab)) = ab = \phi^{-1}(x)\phi^{-1}(y)$. Composition is associative because function composition is. Thus $\mathrm{Aut}(G)$ is a group. **(b)** First prove the subgroup requirement: $i_g i_h = i_{gh}$, i_e is the identity, and $(i_g)^{-1} = i_{g^{-1}}$, so $\mathrm{Inn}(G) = \{i_g : g \in G\}$ is a subgroup. For any $\phi \in \mathrm{Aut}(G)$ and $x \in G$,

$$(\phi i_g \phi^{-1})(x) = \phi(g\phi^{-1}(x)g^{-1}) = \phi(g)x\phi(g)^{-1} = i_{\phi(g)}(x).$$

Thus conjugation by every automorphism sends inner automorphisms to inner automorphisms, proving $\mathrm{Inn}(G) \triangleleft \mathrm{Aut}(G)$.

Exercise 14.38: Elements inducing the identity inner automorphism

The condition $i_g = \mathrm{id}$ means $gxg^{-1} = x$ for every $x \in G$, equivalent to $gx = xg$ for every x. Thus the set is the center $Z(G)$. It is normal because it is the kernel of the homomorphism

$$G \to \mathrm{Aut}(G), \qquad g \mapsto i_g,$$

whose homomorphism law is $i_g i_h = i_{gh}$. Alternatively, conjugating a central element leaves it unchanged, directly giving normality.

Exercise 14.39: An induced map on factor groups

Define $\phi_* : G/H \to G'/H'$ by $\phi_*(gH) = \phi(g)H'$.

1. Check representatives.

If $gH = kH$, then $k^{-1}g \in H$. By the assumption $\phi[H] \subseteq H'$, its image $\phi(k)^{-1}\phi(g)$ is in H'. Thus $\phi(g)H' = \phi(k)H'$ and the proposed map is well defined.

2. Check products.

Normality makes both quotient products valid, and

$$\phi_*((gH)(kH)) = \phi(gk)H' = \phi(g)\phi(k)H' = (\phi(g)H')(\phi(k)H').$$

Therefore ϕ_* is a homomorphism. It is natural because it sends a coset to the coset of the original image without any arbitrary choices.

Exercise 14.40: Normal subgroups defined by determinants

(a) The determinant is a homomorphism $\det : GL(n,\mathbb{R}) \to \mathbb{R}^*$ because $\det(AB) = \det A \det B$ and invertibility gives nonzero determinant. The determinant-1 matrices form its kernel, hence a normal subgroup. Explicitly products and inverses keep determinant 1, and $\det(BAB^{-1}) = \det A$. **(b)** The map $A \mapsto |\det A|$ is a homomorphism to $\mathbb{R}^+$, because absolute value is multiplicative. Its kernel is exactly the matrices with determinant $+1$ or -1. Therefore this set is also a normal subgroup. The identity belongs to both sets because $\det I_n = 1$.

Exercise 14.41: Multiplication of subsets and the quotient group

(a) The set product is associative: $(AB)C$ and $A(BC)$ both consist of all abc with $a \in A, b \in B, c \in C$, since multiplication in G is associative. The identity subset is $\{e\}$, because $\{e\}A = A\{e\} = A$. But the powerset is not a group: the empty set has product $\varnothing$ with every subset and cannot have product $\{e\}$ with an inverse. **(b)** If $N \triangleleft G$, then for $a, b \in G$,

$$(aN)(bN) = \{an_1bn_2 : n_1, n_2 \in N\} = abN.$$

For inclusion in abN, rewrite $an_1bn_2 = ab(b^{-1}n_1b)n_2$, whose final factor lies in N. Conversely every abn is the product of $a \in aN$ and $bn \in bN$, so equality holds. Thus cosets are closed under the subset product, and the rule agrees with representative multiplication. **(c)** Associativity is inherited from subset multiplication. The subset N is an identity on cosets, since $N(aN) = aN = (aN)N$ by the displayed equality and normality. Each aN has inverse $a^{-1}N$, because their products in both orders equal N. Hence the cosets form a group, proved directly from subsets. Its identity is N, whereas the identity of the whole powerset monoid is $\{e\}$; they coincide exactly when $N = \{e\}$.

15 Factor-Group Computations and Simple Groups

Exercise 15.1: Removing an entire direct factor

The projection $(a, b) \mapsto a$ from $\mathbb{Z}_2 \times \mathbb{Z}_4$ onto $\mathbb{Z}_2$ has kernel $\{0\} \times \mathbb{Z}_4 = \langle (0, 1) \rangle$. The first isomorphism theorem therefore identifies the quotient with $\mathbb{Z}_2$.

Exercise 15.2: Reducing the second coordinate modulo two

The map $(a, b) \mapsto (a, b \bmod 2)$ is a surjective homomorphism onto $\mathbb{Z}_2 \times \mathbb{Z}_2$. Its kernel is $\{(0, 0), (0, 2)\} = \langle (0, 2) \rangle$. Hence this is the quotient's classification.

Exercise 15.3: A diagonal relation changes the quotient

Define $\phi : \mathbb{Z}_2 \times \mathbb{Z}_4 \to \mathbb{Z}_4$ by $\phi(a, b) = 2a + b \pmod 4$. It is well defined because changing a by 2 changes the result by 4, and it is onto since $\phi(0, 1) = 1$. Its kernel is exactly $(0, 0), (1, 2)$. Thus the quotient by $\langle (1, 2) \rangle$ is $\mathbb{Z}_4$, rather than $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Exercise 15.4: A cyclic quotient of order eight

The surjective map $(a, b) \mapsto b - 2a \pmod 8$ from $\mathbb{Z}_4 \times \mathbb{Z}_8$ to $\mathbb{Z}_8$ is well defined. Its kernel consists exactly of $(a, 2a)$, the subgroup generated by $(1, 2)$. Therefore the quotient is $\mathbb{Z}_8$. Its order is also $32/4 = 8$, checking the computation.

Exercise 15.5: Eliminating one coordinate in a triple product

Map $\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_8$ onto $\mathbb{Z}_4 \times \mathbb{Z}_8$ by

$$\phi(a, b, c) = (b - 2a \pmod 4,\ c - 4a \pmod 8).$$

Changing the representative for a by 4 adds multiples of 4 and 8 to the outputs, so the formula is well defined. It is onto using $a = 0$. Its kernel is $(a, 2a, 4a)$, precisely $\langle (1, 2, 4) \rangle$. Thus the quotient is $\mathbb{Z}_4 \times \mathbb{Z}_8$, of order $128/4 = 32$.

Exercise 15.6: An infinite coordinate projection

The map $\mathbb{Z}^2 \to \mathbb{Z}$, $(a, b) \mapsto a$, is onto with kernel $\langle (0, 1) \rangle$. Hence the quotient is the infinite cyclic group $\mathbb{Z}$.

Exercise 15.7: A primitive diagonal in the integer lattice

Use $\phi(a, b) = b - 2a$. This homomorphism onto $\mathbb{Z}$ has kernel the pairs $(a, 2a)$, exactly $\langle (1, 2) \rangle$. Thus the quotient is $\mathbb{Z}$, with no finite torsion factor.

Exercise 15.8: A primitive diagonal in three dimensions

The map $\phi : \mathbb{Z}^3 \to \mathbb{Z}^2$ given by $(a, b, c) \mapsto (b - a, c - a)$ is onto and has kernel the triples (a, a, a). Therefore the quotient by $\langle (1, 1, 1) \rangle$ is $\mathbb{Z} \times \mathbb{Z}$.

Exercise 15.9: A finite relation on an infinite coordinate

Reducing the first integer coordinate modulo 3 gives the quotient $\mathbb{Z}_3 \times \mathbb{Z} \times \mathbb{Z}_4$. Since 3 and 4 are coprime, $\mathbb{Z}_3 \times \mathbb{Z}_4 \cong \mathbb{Z}_{12}$ by the Chinese remainder theorem. In invariant-factor form the answer is $\mathbb{Z} \times \mathbb{Z}_{12}$. The middle free coordinate remains infinite.

Exercise 15.10: A fourth-multiple relation

The relation only reduces the second integer coordinate modulo 4. The map $(a, b, c) \mapsto (a, b \bmod 4, c)$ is onto $\mathbb{Z} \times \mathbb{Z}_4 \times \mathbb{Z}_8$ with kernel $\langle (0, 4, 0) \rangle$. Thus that direct product is the classification, and $4 \mid 8$ puts the finite factors in invariant-factor order.

Exercise 15.11: A nonprimitive diagonal

Map $\mathbb{Z}^2$ onto $\mathbb{Z} \times \mathbb{Z}_2$ by $\phi(a, b) = (a - b, b \bmod 2)$. Given $(u, \bar{v})$, choose an integer representative v and use $(u + v, v)$, proving surjectivity. The kernel requires $a = b$ and b even, hence is $\langle (2, 2) \rangle$. Therefore the quotient is $\mathbb{Z} \times \mathbb{Z}_2$. The factor of 2 in the diagonal produces the torsion.

Exercise 15.12: A triple diagonal with torsion

The map $(a, b, c) \mapsto (b - a, c - a, a \bmod 3)$ is onto $\mathbb{Z}^2 \times \mathbb{Z}_3$. Its kernel consists of $a = b = c$ with $3 \mid a$, exactly $\langle (3, 3, 3) \rangle$. Thus the quotient is $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}_3$.

Exercise 15.13: Center and commutators of the square group

Write $D_4 = \langle r, s \rangle$ with $r^4 = s^2 = e$ and $srs = r^{-1}$. A rotation r^k commutes with s exactly when $r^k = r^{-k}$, meaning k even; no reflection commutes with r, since that would give $r = r^{-1}$. Thus $Z(D_4) = \{e, r^2\}$. The commutator $rsr^{-1}s^{-1} = r^2$ lies in C. On the other hand the quotient by $\langle r^2 \rangle$ has $r^{-1} = r$, so its defining generators commute and it is abelian. The commutator subgroup is therefore contained in $\langle r^2 \rangle$ as well, proving $C = \{e, r^2\}$.

Exercise 15.14: A direct product with a symmetric group

For any two groups, the center of their product is the product of their centers: commuting is checked in each coordinate. Likewise $[G \times H, G \times H] = [G, G] \times [H, H]$, since commutators are coordinatewise and commutators in either individual coordinate are obtained by setting the other coordinate to the identity. As $\mathbb{Z}_3$ is abelian, $Z(S_3) = \{e\}$, and $[S_3, S_3] = A_3$, the answers are

$$Z(\mathbb{Z}_3 \times S_3) = \mathbb{Z}_3 \times \{e\}, \qquad C = \{0\} \times A_3 \cong \mathbb{Z}_3.$$

For the last standard fact, S_3/A_3 is abelian, while the commutator of two different transpositions is a nonidentity three-cycle, generating A_3.

Exercise 15.15: Two nonabelian direct factors

Apply the coordinatewise center and commutator rules proved in Exercise 15.14. Since $Z(S_3) = \{e\}$, $Z(D_4) = \{e, r^2\}$, $[S_3, S_3] = A_3$, and $[D_4, D_4] = \langle r^2 \rangle$,

$$Z(S_3 \times D_4) = \{e\} \times \langle r^2 \rangle \cong \mathbb{Z}_2, \qquad C = A_3 \times \langle r^2 \rangle \cong \mathbb{Z}_6.$$

The last isomorphism uses the coprime orders 3 and 2.

Exercise 15.16: Every small subgroup of a sixteen-element group

Let $G = \mathbb{Z}_4 \times \mathbb{Z}_4$. Its complete list of subgroups of order at most 4, with quotients, is

subgroup	order	G/H
$\{(0,0)\}$	1	$\mathbb{Z}_4 \times \mathbb{Z}_4$
$\langle (2,0) \rangle$	2	$\mathbb{Z}_2 \times \mathbb{Z}_4$
$\langle (0,2) \rangle$	2	$\mathbb{Z}_2 \times \mathbb{Z}_4$
$\langle (2,2) \rangle$	2	$\mathbb{Z}_2 \times \mathbb{Z}_4$
$\langle (1,0) \rangle,\ \langle (0,1) \rangle$	4	$\mathbb{Z}_4$ (each)
$\langle (1,1) \rangle,\ \langle (1,2) \rangle$	4	$\mathbb{Z}_4$ (each)
$\langle (1,3) \rangle,\ \langle (2,1) \rangle$	4	$\mathbb{Z}_4$ (each)
$\{(0,0), (2,0), (0,2), (2,2)\}$	4	$\mathbb{Z}_2 \times \mathbb{Z}_2$

Here each bracket denotes all multiples of its displayed generator, so the actual subgroup elements are explicit from that rule. To verify completeness, exactly three nonzero elements have order 2. The remaining twelve nonzero elements have order 4, with two generators per cyclic subgroup, giving exactly six cyclic subgroups as listed. A noncyclic order-four subgroup must consist of the identity and all three order-two elements, so there is exactly one more. For an order-two subgroup, the quotient has order 8, exponent at most 4, and an element of order 4: use a coordinate not killed by that subgroup, or $(1,0) + H$ for $H = \langle (2,2) \rangle$. Its classification is therefore $\mathbb{Z}_2 \times \mathbb{Z}_4$. Each cyclic order-four generator has an odd coordinate; elementary invertible changes of coordinates send it to $(1,0)$, so its quotient is $\mathbb{Z}_4$. The remaining subgroup is $2G$, and reducing both coordinates modulo 2 gives its quotient $\mathbb{Z}_2^2$.

Exercise 15.17: The center must contain exactly the commuting elements

Replace "contains all" by "is the set of all." Precisely,

$$Z(G) = \{z \in G : zg = gz \text{ for every } g \in G\}.$$

Saying only that it contains these elements would allow extraneous elements and would not determine a subgroup uniquely.

Exercise 15.18: The commutator subgroup requires generation

The commutator subgroup is the subgroup *generated* by all commutators, not merely their set. It is

$$C = \langle a^{-1}b^{-1}ab : a, b \in G \rangle.$$

This convention gives the same subgroup as $aba^{-1}b^{-1}$, by replacing a, b with their inverses. Individual commutators need not themselves be closed under multiplication, so generation is essential.

Exercise 15.19: Ten statements about factor and simple groups

(a) True. If a generates G, then aN generates every quotient G/N.

(b) False. $\mathbb{Z}_2^2$ is not cyclic, but quotienting by one coordinate gives $\mathbb{Z}_2$.

(c) False. $1/2 + \mathbb{Z}$ has order 2.

(d) True. $1/n + \mathbb{Z}$ has order n, including the identity when $n = 1$.

(e) False. An element killed by 4 must have representative $k/4$. Modulo integers there are only four such classes, and exactly two have order 4: $1/4 + \mathbb{Z}, 3/4 + \mathbb{Z}$.

(f) True. If every commutator is identity, $aba^{-1}b^{-1} = e$ gives $ab = ba$.

(g) False. The inclusion is $C \subseteq H$, because all commutators become identity in an abelian quotient.

(h) False. A simple cyclic group of prime order is abelian and has trivial commutator subgroup.

(i) True. C is normal; it is nontrivial in a nonabelian group, so simplicity forces $C = G$.

(j) False. A_5 is simple and has order 60, which is not prime.

Exercise 15.20: Functions modulo constants

Let $H = \{f : \mathbb{R} \to \mathbb{R} : f(0) = 0\}$ under addition. Define $T : F \to H$ by $T(f)(x) = f(x) - f(0)$. This is an additive homomorphism and is onto because it fixes every function in H. Its kernel is exactly the constant functions K. Thus $F/K \cong H$. Choosing a basepoint turns each coset into its unique representative vanishing there.

Exercise 15.21: Nonvanishing functions modulo constant factors

Let $H^* = \{f : \mathbb{R} \to \mathbb{R} \setminus \{0\} : f(0) = 1\}$ under multiplication. The map $T(f)(x) = f(x)/f(0)$ is well defined and multiplicative, and fixes every function in H^*, so is onto. Its kernel consists exactly of the nonzero constant functions. Therefore $F^*/K^* \cong H^*$.

Exercise 15.22: No order-two additive coset modulo continuous functions

No. If $f + K$ were killed by 2, then $2f$ would be continuous. Multiplying it by the constant $1/2$ would make f continuous as well, so $f \in K$ and its coset is identity, not an element of order 2. The same argument shows the additive quotient has no nonidentity element of any finite order.

Exercise 15.23: An order-two multiplicative coset

Yes. Define $f(x) = 1$ for $x \geq 0$ and $f(x) = -1$ for $x < 0$. It never vanishes but is discontinuous at 0, so $f \notin K^*$. Yet f^2 is the constant function 1 and lies in K^*. Thus fK^* is nonidentity and its square is identity, giving order exactly 2. Unlike scalar multiplication, squaring loses the discontinuous sign information.

Exercise 15.24: A translated full circle

Since $z_0 \in U$, multiplication by z_0 maps U onto U: for any $w \in U$, $w = z_0(z_0^{-1}w)$ and $z_0^{-1}w \in U$. Hence $z_0U = U$, which is certainly a subgroup. Its quotient $U/z_0U = U/U$ is the trivial group.

Exercise 15.25: The circle modulo a half-turn

The homomorphism $U \to U$, $z \mapsto z^2$, is onto because every unit complex number has a unit square root. Its kernel is $\{1, -1\} = \langle -1 \rangle$. Thus $U/\langle -1 \rangle \cong U$ by the first isomorphism theorem.

Exercise 15.26: The circle modulo the nth roots

The map $U \to U$, $z \mapsto z^n$, is a surjective homomorphism. Its kernel is the cyclic subgroup of all nth roots of unity, generated by ζ_n. Consequently $U/\langle \zeta_n \rangle \cong U$. When $n = 1$ this is the identity-map case.

Exercise 15.27: Real numbers modulo integers

The map $\phi : \mathbb{R} \to U$ given by $\phi(t) = \cos(2\pi t) + i \sin(2\pi t)$ is a homomorphism from addition to multiplication, by the angle-addition formulas. It is onto, and $\phi(t) = 1$ exactly when $t \in \mathbb{Z}$. Thus $\mathbb{R}/\mathbb{Z} \cong U$.

Exercise 15.28: Torsion created by quotienting

Take the torsion-free additive group $G = \mathbb{Z}$ and $H = 2\mathbb{Z}$. In G, $na = 0$ with $n > 0$ forces $a = 0$, so no nonzero element has finite order. But $G/H \cong \mathbb{Z}_2$, and all its elements have finite order. Alternatively $\mathbb{Q}/\mathbb{Z}$ gives an infinite quotient with the same property.

Exercise 15.29: Isomorphic kernels with different quotients

Let $G = \mathbb{Z} \times \mathbb{Z}_2$, $H = \mathbb{Z} \times \{0\}$, and $K = 2\mathbb{Z} \times \{0\}$. They are normal because G is abelian, and both are infinite cyclic, hence isomorphic. However $G/H \cong \mathbb{Z}_2$, while $G/K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Their different orders show the quotients are not isomorphic.

Exercise 15.30: Centers of simple groups

(a) If a simple group is abelian, every element commutes with every other, so $Z(G) = G$. A nontrivial simple abelian group is cyclic of prime order.

(b) If it is nonabelian, its normal center cannot be G. Simplicity leaves only $Z(G) = \{e\}$.

Exercise 15.31: Commutators of simple groups

(a) In an abelian group, every commutator is e, so $C = \{e\}$.

(b) In a nonabelian group at least one commutator is nonidentity. Thus C is a nontrivial normal subgroup, and simplicity forces $C = G$.

Exercise 15.32: Synopsis of cyclic quotients

If $G = \langle a \rangle$, then every coset is $a^k N = (aN)^k$, so aN generates G/N. This one observation proves the cyclic-quotient theorem.

Exercise 15.33: Synopsis of maximal normal subgroups

Under the quotient map $G \to G/M$, normal subgroups properly between M and G correspond exactly to nontrivial proper normal subgroups of G/M. Thus absence of the former, meaning M is maximal normal, is equivalent to simplicity of G/M.

Exercise 15.34: Index two prevents simplicity

A subgroup H of index 2 is normal: for $g \notin H$, both gH and Hg are the complement of H, while for $g \in H$ both equal H. The subgroup is proper because its index is 2 and nontrivial by hypothesis. It is therefore a nontrivial proper normal subgroup of G, so G is not simple.

Exercise 15.35: The image of a normal subgroup

The image $\phi[N]$ is a subgroup of $\phi[G]$. For $\phi(g) \in \phi[G]$ and $\phi(n) \in \phi[N]$,

$$\phi(g)\phi(n)\phi(g)^{-1} = \phi(gng^{-1}) \in \phi[N],$$

since N is normal in G. Thus the image is normal in $\phi[G]$. It need not be normal in all of G' when ϕ is not onto.

Exercise 15.36: The inverse image of a normal subgroup

Set $H = \phi^{-1}[N']$. It is a subgroup because identity, products and inverses map into the subgroup N'. For $g \in G, h \in H$,

$$\phi(ghg^{-1}) = \phi(g)\phi(h)\phi(g)^{-1} \in N',$$

by normality of N' in G'. Hence $ghg^{-1} \in H$, proving normality in G. No surjectivity assumption is needed.

Exercise 15.37: A cyclic central quotient forces commutativity

Prove the contrapositive. Suppose $G/Z(G)$ is generated by $aZ(G)$. Each $x \in G$ then has form $a^m z$ with $z \in Z(G)$, and similarly $y = a^n w$. Since the central factors commute with all elements,

$$xy = a^{m+n} zw = a^{n+m} wz = yx.$$

Thus G is abelian. Therefore a nonabelian group cannot have a cyclic quotient by its center.

Exercise 15.38: A nonabelian group of order a product of primes

The center is a subgroup, so its order divides pq. If it were the whole group, G would be abelian. If it were nontrivial and proper, its order would be p or q (also when $p = q$). The quotient would then have prime order and hence be cyclic. Exercise 15.37 would make G abelian, again a contradiction. Thus $Z(G) = \{e\}$.

Exercise 15.39: A complete proof of simplicity of the alternating group

We compose permutations from right to left. All letters appearing distinctly in a displayed identity are assumed distinct. Put $c_i = (r, s, i)$ for fixed distinct r, s and $i \notin \{r, s\}$. The source's description of "n special cycles" means these $n - 2$ actual three-cycles; repeated entries $i = r, s$ are not cycles.

(a) A three-cycle is a product of two transpositions, for example $(a, b, c) = (a, c)(a, b)$, so is even and lies in A_n.

(b) Every even permutation is a product of an even number of transpositions. Group them into adjacent pairs. Equal transpositions cancel. A pair sharing one letter is a three-cycle (or its inverse). For a disjoint pair, direct computation gives $(a, b)(c, d) = (a, c, b)(a, c, d)$. Thus every such pair is a product of three-cycles, and these generate A_n.

(c) To show the special three-cycles suffice, compute

$$c_i^2 = (r, i, s), \qquad c_j c_i^2 = (r, i, j),$$
$$c_j^2 c_i = (s, i, j), \qquad c_i^2 c_k c_j^2 c_i = (i, j, k).$$

Their inverses give the opposite orientations. These cover three-cycles containing both r, s, only r, only s, or neither. Thus every three-cycle is a product of the special ones, and (b) proves generation. If $n = 3$, only c_i, c_i^{-1} are needed.

(d) Suppose $N \trianglelefteq A_n$ contains a three-cycle, relabeled as c_i. Then $c_i^2 \in N$. For any other $j \notin \{r, s, i\}$, the double transposition $h = (r, s)(i, j)$ is even, and conjugation gives $hc_i^2 h^{-1} = (r, s, j) = c_j \in N$. Thus N contains all special cycles (and c_i itself). Part (c) gives $N = A_n$. For $n = 3$ the contained three-cycle already generates A_3.

(e) Take a nonidentity $\sigma \in N$ and use its disjoint-cycle decomposition. We check every case in the source. For any $t \in A_n$, normality gives $\sigma^{-1} t\sigma t^{-1} \in N$: the first factor σ^{-1} and the conjugate $t\sigma t^{-1}$ both belong to N.

Case I: N contains a three-cycle as an element.

Part (d) applies immediately. Merely having a three-cycle among the disjoint factors of a larger permutation does not suffice; those cases are treated below.

Case II: there is a cycle of length $\ell > 3$.

Write $\sigma = \mu(a_1, a_2, \ldots, a_\ell)$, with μ disjoint from that cycle. Set $t = (a_1, a_2, a_3)$. Conjugation replaces the entries by their preimages under σ, so

$$\sigma^{-1}t\sigma t^{-1} = (a_\ell, a_1, a_2)(a_1, a_3, a_2) = (a_1, a_3, a_\ell).$$

This is a three-cycle in N, so (d) applies.

Case III: at least two three-cycles occur.

Write $\sigma = \mu(a_4, a_5, a_6)(a_1, a_2, a_3)$, all factors disjoint, and choose $t = (a_1, a_2, a_4)$. Then

$$\sigma^{-1}t\sigma t^{-1} = (a_3, a_1, a_6)(a_1, a_4, a_2) = (a_1, a_4, a_2, a_6, a_3).$$

This five-cycle lies in N, reducing to Case II, hence to Case I.

Case IV: precisely one three-cycle and some two-cycles occur.

Here $\sigma = \mu(a_1, a_2, a_3)$ with μ a disjoint product of transpositions. Since $\mu^2 = e$ and disjoint factors commute, $\sigma^2 = (a_1, a_3, a_2) \in N$, a three-cycle.

Case V: all nontrivial cycles are transpositions.

Since σ is even, there are an even number of them, at least two. Write $\sigma = \mu(a_3, a_4)(a_1, a_2)$ with μ disjoint. Taking $t = (a_1, a_2, a_3)$ gives

$$\sigma^{-1}t\sigma t^{-1} = (a_2, a_1, a_4)(a_1, a_3, a_2) = (a_1, a_3)(a_2, a_4) = \alpha \in N.$$

Because $n \geq 5$, choose a letter j different from these four and let $\beta = (a_1, a_3, j) \in A_n$. Another conjugation calculation gives

$$\beta^{-1}\alpha\beta\alpha = (j, a_1)(a_2, a_4)(a_1, a_3)(a_2, a_4) = (a_1, a_3, j) \in N.$$

This is again a three-cycle. These cases exhaust all possible nonidentity even permutations: a longer cycle gives II; otherwise the number of three-cycles is at least two, exactly one, or zero, giving III, IV, or V. Every nontrivial normal subgroup is therefore A_n, proving simplicity for $n \geq 5$.

Exercise 15.40: The product of a subgroup and a normal subgroup

The set HN contains both subgroups. For $h_1n_1, h_2n_2 \in HN$, normality gives

$$(h_1n_1)(h_2n_2) = h_1h_2(h_2^{-1}n_1h_2)n_2 \in HN.$$

Also $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}(hn^{-1}h^{-1}) \in HN$. Together with the identity this proves HN is a subgroup. Any subgroup containing both H and N contains every product hn, so contains HN. Thus HN is the smallest such subgroup.

Exercise 15.41: A product of two normal subgroups

By Exercise 15.40, NM is a subgroup. If $g \in G$ and $nm \in NM$, then $g(nm)g^{-1} = (gng^{-1})(gmg^{-1})$. The two factors lie in N and M, respectively, by their normality, so the product lies in NM. Therefore NM is normal in G.

Exercise 15.42: Disjoint normal subgroups commute elementwise

For $h \in H, k \in K$, let $c = hkh^{-1}k^{-1}$. It belongs to K, because $hkh^{-1} \in K$ and $k^{-1} \in K$. It also belongs to H, because $kh^{-1}k^{-1} \in H$ and $c = h(kh^{-1}k^{-1})$. The intersection hypothesis gives $c = e$, so $hk = kh$. Thus all pairs of elements from the two subgroups commute.

16 Group Action on a Set

For a group action, gx denotes the result of applying g to x; it need not be multiplication in a group containing x. The axioms are $ex = x$ and $(gh)x = g(hx)$. The orbit Gx consists of all possible images of x, the stabilizer G_x consists of the group elements fixing x, and X_g is the set of points fixed by one specified g. Keep these three sets distinct. In the classification exercises, a transitive set is understood to be nonempty, as in the book's examples. If the empty set is admitted, the quantified transitivity condition holds vacuously there and adds one empty case.

Exercise 16.1: All fixed sets in the square action

Plan. In each row of Table 16.10, retain exactly the column labels that the row leaves unchanged. Write X for the full seventeen-element set.

$$X_{\rho_0} = X, \qquad X_{\rho_1} = \{C\}, \qquad X_{\rho_3} = \{C\},$$
$$X_{\rho_2} = \{m_1, m_2, d_1, d_2, C\},$$
$$X_{\mu_1} = \{s_1, s_3, m_1, m_2, C, P_1, P_3\},$$
$$X_{\mu_2} = \{s_2, s_4, m_1, m_2, C, P_2, P_4\},$$
$$X_{\delta_1} = \{2, 4, d_1, d_2, C\},$$
$$X_{\delta_2} = \{1, 3, d_1, d_2, C\}.$$

For example, a half-turn exchanges opposite vertices, sides, and midpoints, but leaves each axis and each diagonal as a set. An object in X can be a whole line or side, so "fixed" does not require each point of that object to be fixed individually.

Exercise 16.2: Every stabilizer in the square action

Plan. Read down a column and collect the row labels that reproduce its heading. Let $e = \rho_0$. The answers can be grouped by identical stabilizers:

x	G_x
$1, 3$	$\{e, \delta_2\}$
$2, 4$	$\{e, \delta_1\}$
s_1, s_3, P_1, P_3	$\{e, \mu_1\}$
s_2, s_4, P_2, P_4	$\{e, \mu_2\}$
m_1, m_2	$\{e, \rho_2, \mu_1, \mu_2\}$
d_1, d_2	$\{e, \rho_2, \delta_1, \delta_2\}$
C	D_4

For example, vertex 1 is fixed by the identity and by reflection in its own diagonal d_2, and by no other square symmetry. The orbit–stabilizer check gives orbit size $8/2 = 4$, as expected for the vertices.

Exercise 16.3: The six orbits of the square action

Starting from a vertex reaches every vertex and no other kind of object. The same observation for each kind of object gives

$$\{1, 2, 3, 4\}, \quad \{s_1, s_2, s_3, s_4\}, \quad \{m_1, m_2\}, \quad \{d_1, d_2\}, \quad \{C\}, \quad \{P_1, P_2, P_3, P_4\}.$$

They are disjoint, their sizes sum to 17, and each is closed under all square symmetries. This verifies that the list exhausts X.

Exercise 16.4: Correcting faithful action

The quantifier "for every x" is essential. An action is faithful if

$$(gx = x \text{ for every } x \in X) \implies g = e.$$

Equivalently, the homomorphism $G \to S_X$ determined by the action is injective. Requiring this implication for a single fixed point would mean that all point stabilizers are trivial, which is the stronger condition called a free action. A reflection of a square fixes some vertices without acting as the identity on all vertices.

Exercise 16.5: Correcting transitive action

For every $x, y \in X$ there must exist $g \in G$ such that $gx = y$. The group element may depend on both points. One fixed g cannot send a single x to several different outputs. For a nonempty X, an equivalent formulation is: the orbit of any one chosen point is all of X.

Exercise 16.6: Subsets preserved by an action

Conclusion. The sub-G-sets are precisely the unions of whole G-orbits.

If S is a sub-G-set and $s \in S$, closure gives $Gs \subseteq S$. Since $s = es$, every s is in its own orbit, so $S = \bigcup_{s \in S} Gs$. Conversely, if S is a union of orbits and $s \in Gx \subseteq S$, then $s = hx$ for some h. For any g, $gs = (gh)x \in Gx \subseteq S$. The empty union also yields the empty sub-G-set.

Exercise 16.7: Transitivity in terms of orbits

For nonempty X, transitivity is equivalent to having exactly one orbit. If every y equals gx for some g, then $Gx = X$. Conversely, when $Gx = X$, any $y = hx$ and $z = kx$ satisfy $(kh^{-1})y = z$. If empty sets are allowed under the literal quantified definition, the corresponding characterization is "at most one orbit."

Exercise 16.8: True or false: action properties

(a) False. A group can act on any set; a three-element set with the trivial $\mathbb{Z}_2$ action need not come equipped with a group operation.

(b) True. This is the axiom $ex = x$.

(c) False. In the trivial action every g fixes every point; faithfulness is an additional condition.

(d) True. Apply g^{-1} to $gx_1 = gx_2$ and use the action axioms to get $x_1 = x_2$.

(e) False. In the trivial action of a nontrivial group all group elements send a chosen x to the same point.

(f) True. It is preserved by G, and if hx, kx are two points in the orbit then kh^{-1} sends the first to the second.

(g) True. Restrict the action map to $H \times X$; the same two axioms hold.

(h) False. Take $H = \{e\}$ inside a group that acts transitively on more than one point. Its orbits are singletons, while the original orbit is larger.

(i) True. The inverse of the map $x \mapsto gx$ is $x \mapsto g^{-1}x$.

(j) True. Orbit–stabilizer gives $|Gx| = [G : G_x] = |G|/|G_x|$ when G is finite.

Exercise 16.9: Isomorphic orbits and other subactions

(a) The side orbit and midpoint orbit are isomorphic by $s_i \mapsto P_i$. Every square symmetry sends the midpoint of a side to the midpoint of its image side, so this bijection commutes with the action.

(b) The reflection μ_1 fixes two sides, s_1, s_3, but no vertices. An equivariant bijection must carry the fixed points of any g bijectively onto the fixed points of the same g. Thus no such bijection can identify the vertex and side orbits.

(c) No. The question permits sub-D_4-sets that are unions of orbits. For instance $\{s_1, s_2, s_3, s_4, C\}$ and $\{P_1, P_2, P_3, P_4, C\}$ are different isomorphic sub-D_4-sets: extend the bijection in part (a) by $C \mapsto C$. Among individual orbits, the only isomorphic pair is indeed the side and midpoint pair: the other four-point orbit is excluded by part (b), while μ_1 fixes both axes and exchanges the diagonals, distinguishing the two two-point orbits.

Exercise 16.10: Faithfulness of the square action

(a) Yes. A symmetry fixing all seventeen objects fixes all four vertices. Only the identity symmetry does that.

(b) The vertex, side, and midpoint orbits each give faithful actions: inspection of their columns in Table 16.10 shows that no nonidentity row fixes the entire orbit. The axis orbit and diagonal orbit are not faithful, since the nonidentity half-turn ρ_2 fixes both members in either case. The center orbit is not faithful, since every symmetry fixes C.

Exercise 16.11: Faithfulness and distinct permutations

What we must show. Prove that faithfulness says precisely that different group elements induce different maps on X. Suppose first that the action is faithful and $gx = hx$ for every x. Applying h^{-1} gives $(h^{-1}g)x = x$ for all x, so $h^{-1}g = e$ and $g = h$. Conversely, if distinct group elements have distinct actions and $gx = x$ for every x, then g and e induce the same map. Hence $g = e$, proving faithfulness. Equality of actions here means equality on *every* point, not merely agreement at one point.

Exercise 16.12: Pointwise stabilizers of subsets

Let $G_Y = \{g : gy = y \text{ for every } y \in Y\}$. The identity belongs to G_Y. If $g, h \in G_Y$, then for every $y \in Y$, $(gh)y = g(hy) = gy = y$, so $gh \in G_Y$. If $g \in G_Y$, applying g^{-1} to $gy = y$ gives $g^{-1}y = y$, so $g^{-1} \in G_Y$. Thus G_Y is a subgroup. Equivalently $G_Y = \bigcap_{y \in Y} G_y$. When $Y = \varnothing$, all conditions are vacuous and $G_Y = G$.

Exercise 16.13: The real rotation action

(a) For $\theta \in (\mathbb{R}, +)$ use $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. We have $R_0 = I$ and the angle-addition identities give $R_{\theta+\phi} = R_\theta R_\phi$. These are the two action axioms.

(b) Rotation preserves distance to the origin. Conversely every point at distance $|P| > 0$ has an angle differing from that of P by some real θ. Hence the orbit is the entire circle centered at the origin through P.

(c) A nonzero point returns to itself exactly after an integer multiple of a full turn, so $G_P = 2\pi\mathbb{Z}$ as a subgroup of $(\mathbb{R}, +)$. In particular, this real-number action is not faithful: 2π acts trivially.

Exercise 16.14: Unions of disjoint actions

(a) If the X_i are disjoint, every x in their union belongs to exactly one X_i. Define gx using that component's given action. This is well defined, stays in X_i, and satisfies $ex = x$ and $(gh)x = g(hx)$ because the component action does.

(b) Every $x \in X$ belongs to Gx. Two orbits meeting at a point are equal: if $gx = hy$, then $x = g^{-1}hy$, so every image of x is an image of y, and conversely. Thus the distinct orbits partition X, and their union, with the component actions, is exactly the original action.

Exercise 16.15: A transitive action is a coset action

Put $H = G_{x_0}$ and define $\phi(gx_0) = gH$; transitivity provides such a g for each point.

Prove the definition does not depend on the choice of g.

If $gx_0 = hx_0$, then $(h^{-1}g)x_0 = x_0$, so $h^{-1}g \in H$ and $gH = hH$. The same equivalences in reverse show that $gH = hH$ implies $gx_0 = hx_0$. Thus ϕ is both well defined and injective.

Check surjectivity and the action.

Every coset gH is the image of gx_0, so ϕ is onto. For $k \in G$,

$$\phi(k(gx_0)) = \phi((kg)x_0) = (kg)H = k\phi(gx_0).$$

This proves the required G-set isomorphism.

Exercise 16.16: Decomposing an arbitrary action

First partition X into its distinct orbits O_i. Choose a point x_i in each nonempty orbit and let $H_i = G_{x_i}$. Exercise 16.15 supplies an isomorphism $O_i \to G/H_i$, where G/H_i here denotes the *set of left cosets*; H_i need not be normal. Tag the coset set by i and let

$$g \cdot (kH_i, i) = (gkH_i, i).$$

The map taking $x \in O_i$ to $(\phi_i(x), i)$ is bijective, because the orbits partition X and each ϕ_i is bijective. It is equivariant in each component and hence globally. The tags prevent cosets from different components from being confused. For an arbitrary infinite family, choosing all the representatives uses the usual choice convention; finite actions need no such qualification.

Exercise 16.17: When two coset actions are isomorphic

(a) Let $H = G_{x_0}$. An element k fixes g_0x_0 exactly when $(g_0^{-1}kg_0)x_0 = x_0$, equivalently $g_0^{-1}kg_0 \in H$. Consequently $G_{g_0x_0} = g_0Hg_0^{-1}$.

(b) The conjecture is that G/H and G/K are isomorphic as G-sets exactly when H and K are conjugate subgroups.

(c) If $K = tHt^{-1}$, define $\phi(gH) = gt^{-1}K$. Replacing g by gh for $h \in H$ does not change the answer, because $ght^{-1}K = gt^{-1}(tht^{-1})K = gt^{-1}K$. Its inverse sends uK to utH, and left multiplication commutes with ϕ, proving sufficiency. Conversely, an equivariant bijection sends H to some aK and preserves stabilizers. The stabilizer of H is H, while that of aK is aKa^{-1}. Thus $H = aKa^{-1}$, proving necessity.

Exercise 16.18: All transitive actions of the group of order four

Plan. A transitive action is a coset action, and isomorphism types correspond to conjugacy classes of subgroups. Since $\mathbb{Z}_4$ is abelian, conjugation does not identify different subgroups. Its subgroups are $\mathbb{Z}_4$, $\{0, 2\}$, and $\{0\}$, giving exactly three types, with $1, 2, 4$ points respectively. Their action tables are

	a
0	a
1	a
2	a
3	a

	a	b
0	a	b
1	b	a
2	a	b
3	b	a

	a	b	c	d
0	a	b	c	d
1	b	c	d	a
2	c	d	a	b
3	d	a	b	c

.

In the four-point table, a, b, c, d represent $0, 1, 2, 3$; in the two-point table they represent the two cosets of $\{0, 2\}$.

Exercise 16.19: All transitive actions of the group of order six

There is one subgroup of each order $1, 2, 3, 6$ in $\mathbb{Z}_6$ and no others, since every subgroup of a cyclic group is cyclic. Conjugation is trivial. The four isomorphism types have sizes $6, 3, 2, 1$. For the first

three smaller sizes the tables are

	a
0	a
1	a
2	a
3	a
4	a
5	a

	a	b
0	a	b
1	b	a
2	a	b
3	b	a
4	a	b
5	b	a

	a	b	c
0	a	b	c
1	b	c	a
2	c	a	b
3	a	b	c
4	b	c	a
5	c	a	b

.

The six-point regular action is

	a	b	c	d	e	f
0	a	b	c	d	e	f
1	b	c	d	e	f	a
2	c	d	e	f	a	b
3	d	e	f	a	b	c
4	e	f	a	b	c	d
5	f	a	b	c	d	e

.

Each row adds its group element to the residue represented by the column, modulo the number of points. Since that number divides 6, this defines an action of $\mathbb{Z}_6$ in every case.

Exercise 16.20: All transitive actions of the symmetric group on three letters

Write $r = (1,2,3)$, $r^2 = (1,3,2)$, $s = (2,3)$, $t = (1,3)$, and $u = (1,2)$. The subgroups are S_3, A_3, the three conjugate order-two subgroups, and the identity subgroup. Hence there are four transitive isomorphism types, with $1, 2, 3, 6$ points. The one-point action is trivial; the two-point action records even or odd parity; the three-point action is the natural permutation action; and the six-point action is left multiplication on S_3. Here are all their action tables, with the requested row order:

	a
e	a
r	a
r^2	a
s	a
t	a
u	a

	a	b
e	a	b
r	a	b
r^2	a	b
s	b	a
t	b	a
u	b	a

	a	b	c
e	a	b	c
r	b	c	a
r^2	c	a	b
s	a	c	b
t	c	b	a
u	b	a	c

,

	a	b	c	d	e	f
e	a	b	c	d	e	f
r	b	c	a	f	d	e
r^2	c	a	b	e	f	d
s	d	e	f	a	b	c
t	e	f	d	c	a	b
u	f	d	e	b	c	a

.

In the last table the point labels a, b, c, d, e, f represent the group elements e, r, r^2, s, t, u respectively. For example, $rs = u$, so row r, column d has entry f. The point label e in that table is distinct from the row label for the group identity.

17 Applications of G-Sets to Counting

Counting method. All these exercises request Burnside's formula:

$$\#\text{orbits} = \frac{1}{|G|}\sum_{g\in G}|X_g|.$$

Choose the labeled objects X and the allowed symmetry group G first. A coloring fixed by a permutation must be constant on each cycle of that permutation. With q available colors and c cycles, there are q^c fixed colorings when repetition is allowed. For three-dimensional solids we use rigid rotations; a square piece of cardboard can also be turned over, which induces the plane reflections of its edges.

Exercise 17.1: A cyclic action on eight points

The generator $(1,3,5,6)$ has order four, so the group contains its zeroth, first, second, and third powers. The identity fixes all 8 points. Each of the other three powers moves all four points $1,3,5,6$ and fixes $2,4,7,8$, so each has 4 fixed points. Burnside gives

$$\#\text{orbits} = \frac{8+4+4+4}{4} = \boxed{5}.$$

As a check, the orbits are $\{1,3,5,6\}$ and the four singleton fixed points.

Exercise 17.2: Two generators with disjoint supports

Let $a=(1,3)$ and $b=(2,4,7)$. They commute, have orders 2 and 3, and generate the six distinct elements a^ib^j with $i=0,1$ and $j=0,1,2$. For a point in $\{1,3\}$ to be fixed we need $i=0$; this contributes two fixed points for each such element. A point in $\{2,4,7\}$ is fixed exactly when $j=0$, contributing three. The points $5,6,8$ are always fixed. Thus, in the order e,a,b,b^2,ab,ab^2, the fixed counts are $8,6,5,5,3,3$. Consequently

$$\#\text{orbits} = \frac{8+6+5+5+3+3}{6} = \boxed{5}.$$

Exercise 17.3: Tetrahedral dice with four different labels

The rotation group of a regular tetrahedron has 12 elements: one can choose the face on the bottom in four ways and rotate it into three orientations. There are $4! = 24$ labeled placements of the four different face labels. The identity fixes all 24. Every nonidentity rotation permutes at least two faces, so it cannot fix a placement with distinct labels. Burnside therefore gives

$$\frac{24+11\cdot 0}{12} = \boxed{2}$$

distinguishable dice. Reflections are not rigid rotations of a solid die.

Exercise 17.4: Cube faces with six different colors

There are $8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 20160$ assignments of six distinct colors to six specified faces. The rotation group has order 24. Only the identity can fix a coloring with all faces different, because any other rotation moves some face to another face of a different color. Burnside gives $20160/24 = \boxed{840}$ distinguishable blocks.

Exercise 17.5: Cube faces with arbitrary color repetition

Plan. Divide the 24 rotations into geometric types and count their cycles on the six faces. A fixed coloring is constant on each face cycle.

Rotation type	Number	Face cycle lengths	Cycles
Identity	1	$1,1,1,1,1,1$	6
Quarter-turn about face centers	6	$1,1,4$	3
Half-turn about face centers	3	$1,1,2,2$	4
Third-turn about opposite vertices	8	$3,3$	2
Half-turn about opposite edges	6	$2,2,2$	3

For instance a face-axis quarter-turn fixes the two axial faces and cycles the other four, giving three freely chosen colors. The entries in the second column sum to 24. With $q = 8$, Burnside gives

$$\frac{8^6 + 6 \cdot 8^3 + 3 \cdot 8^4 + 8 \cdot 8^2 + 6 \cdot 8^3}{24} = \frac{281088}{24} = \boxed{11712}.$$

Repeated colors are allowed independently on different cycles.

Exercise 17.6: Coloring the eight vertices of a cube

The same rotations now act on vertices, with different cycle counts:

Rotation type	Number	Vertex cycle lengths	Cycles
Identity	1	$1,1,1,1,1,1,1,1$	8
Face-axis quarter-turn	6	$4,4$	2
Face-axis half-turn	3	$2,2,2,2$	4
Vertex-axis third-turn	8	$1,1,3,3$	4
Edge-axis half-turn	6	$2,2,2,2$	4

For example, a third-turn fixes its two axial vertices and cycles the other six in two triples. With four available colors, Burnside gives

$$\frac{4^8 + 6 \cdot 4^2 + (3+8+6)4^4}{24} = \frac{69984}{24} = \boxed{2916}.$$

Here a color need not appear; if it does, it may appear on any number of vertices. If one additionally requires all four colors to appear, which is a different restriction, replace each 4^c by the number of onto maps from its c cycles to four colors; the result would be 1718.

Exercise 17.7: Painting the edges of a square

The allowed group is D_4, of order 8. Turning over the cardboard makes reflections physically available as well as in-plane rotations.

(a) With no repeated color there are $6 \cdot 5 \cdot 4 \cdot 3 = 360$ labeled paintings. Only the identity fixes any of them. Hence Burnside gives $360/8 = \boxed{45}$.

(b) On the four edges, the identity has four cycles; two quarter-turns have one cycle each; the half-turn has two cycles; two reflections have two cycles; and two reflections have three cycles. Therefore

$$\frac{6^4 + 2 \cdot 6 + (1+2)6^2 + 2 \cdot 6^3}{8} = \frac{1848}{8} = \boxed{231}.$$

A reflection through opposite side midpoints fixes those two edges and swaps the other two, accounting for the three-cycle type.

Exercise 17.8: Resistors on the edges of a tetrahedron

Let X be the 2^6 assignments of two resistor types to the six edges. Use the 12 rigid rotations of the tetrahedral framework. The identity has six edge cycles. Each of the eight third-turns around a vertex and the opposite face center divides the edges into two triples, hence has two cycles. Each of the three half-turns around opposite edge midpoints fixes those two edges as sets and swaps the other four in two pairs, hence has four cycles. Burnside gives

$$\frac{2^6 + 8 \cdot 2^2 + 3 \cdot 2^4}{12} = \frac{144}{12} = \boxed{12}.$$

The objects are actual frameworks in space, so mirror reflection is not an additional rigid rotation. If mirror images were expressly identified, the larger symmetry group would instead give 11 classes.

Exercise 17.9: Coloring a prism with two square ends

Identify the rotation group.

The unique long axis must map to itself, so arbitrary cube rotations cannot be used. There are four rotations about the long axis, through $0, 90, 180, 270$ degrees, and four half-turns about transverse axes. Thus $|G| = 8$. On the six faces the cycle counts are as follows:

Rotation type	Number	Face cycle lengths	Cycles
Identity	1	$1,1,1,1,1,1$	6
Long-axis quarter-turn	2	$1,1,4$	3
Long-axis half-turn	1	$1,1,2,2$	4
Transverse half-turn through side centers	2	$1,1,2,2$	4
Transverse half-turn through side-edge midpoints	2	$2,2,2$	3

The transverse rotations interchange the two square ends. In the first transverse type, two lateral faces remain individually fixed; in the second type the four lateral faces are interchanged in pairs.

(a) With all six colors distinct there are $6! = 720$ labeled colorings, and only the identity fixes any. Burnside gives $720/8 = \boxed{90}$.

(b) With repetition, each cycle can have any of six colors. Hence

$$\frac{6^6 + 3 \cdot 6^4 + 4 \cdot 6^3}{8} = \frac{46656 + 3888 + 864}{8} = \frac{51408}{8} = \boxed{6426}.$$

Answer-key discrepancy. The scanned book's odd-answer key prints 6246 for part (b). The rotation table above gives 6426. Independently, enumerating all 46656 labeled face colorings and identifying them under the eight signed-coordinate rotation matrices preserving the long axis also gives 6426 orbits. Thus the printed answer has transposed digits.